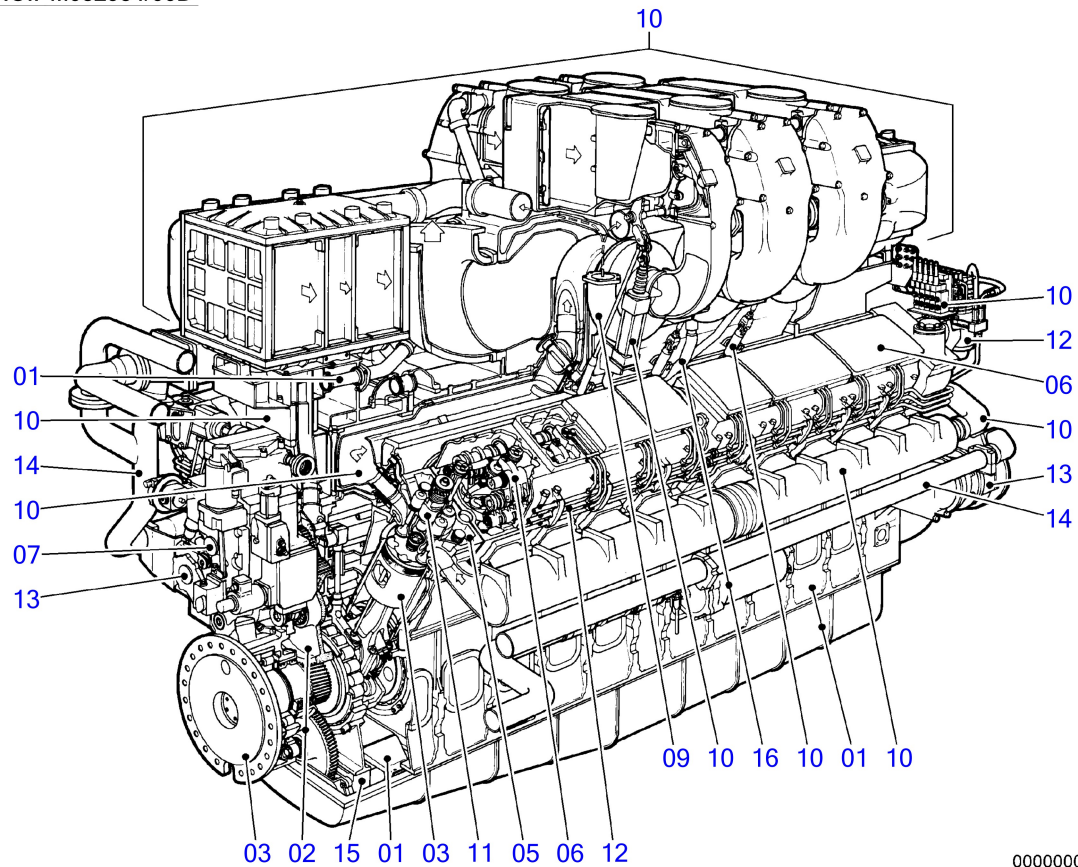




0000002b

FG FG DESIGNATION

- 01 CRANKCASE
- 02 GEAR TRAIN
- 03 RUNNING GEAR
- 04 COVER PLATES AT KGS
- 05 CYLINDER HEAD
- 06 VALVE GEAR
- 07 ENGINE GOVERNOR
- 09 EXHAUST TURBOCHARGER
- 10 CHARGE AIR PIPING-EXHAUST PIPING
- 11 FUEL INJECTION EQUIPMENT
- 12 FUEL SYSTEM
- 13 COOLANT PUMP
- 14 COOLANT SYSTEM
- 15 OIL PUMP
- 16 OIL SYSTEM
- 17 DRIVE FOR SPEED MONITORING
- 18 STARTING EQUIPMENT
- 19 ENGINE MOUNTING
- 84 MONITORING EQUIPMENT
- 86 ACCESSORIES (ELECTRIC, PNEUMATIC)
- 87 ENGINE AND GEARBOX CONTROL



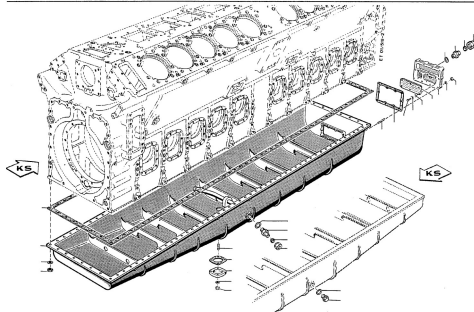
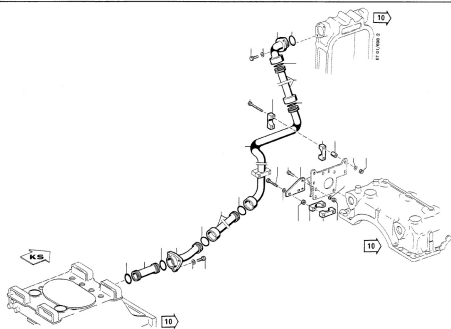
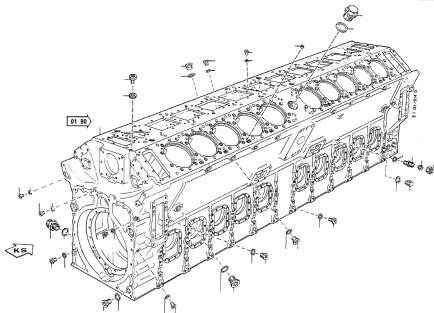
**Warning**  
Do not open hot crankcase for at least  
10 minutes after shutdown of engine.  
Do not restart overheated engine before  
cause of overheating has been remedied.

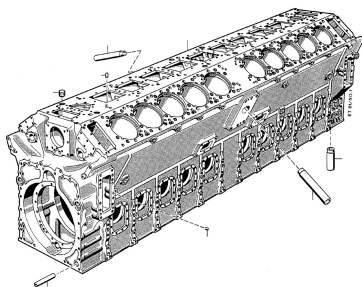
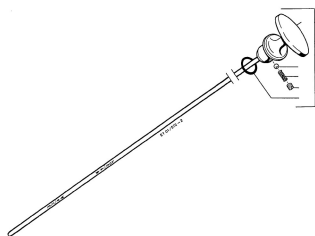
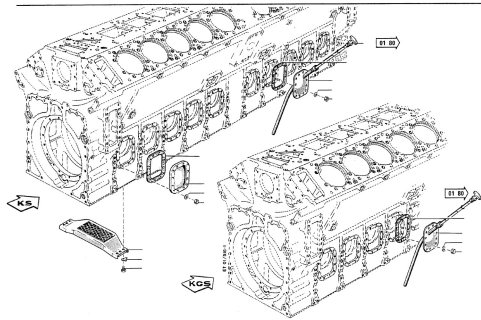
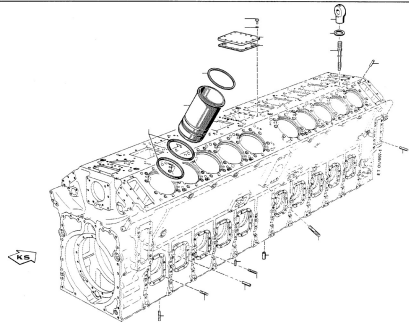
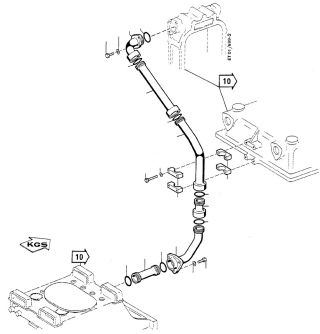
**Hinweise für Betrieb und Wartung**

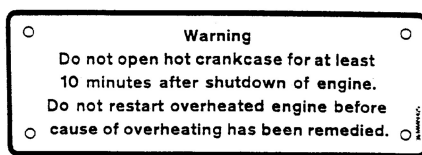
|                   |                                 |                                 |
|-------------------|---------------------------------|---------------------------------|
| Modell            | Modell mit P2                   | Modell                          |
| Leistung          | Leistung bei 1500 min.          | Leistung bei 1800 min.          |
| Umschaltzeitpunkt | Umschaltzeitpunkt bei 1500 min. | Umschaltzeitpunkt bei 1800 min. |
| Umschaltzeitpunkt | Umschaltzeitpunkt bei 1500 min. | Umschaltzeitpunkt bei 1800 min. |
| Umschaltzeitpunkt | Umschaltzeitpunkt bei 1500 min. | Umschaltzeitpunkt bei 1800 min. |
| Umschaltzeitpunkt | Umschaltzeitpunkt bei 1500 min. | Umschaltzeitpunkt bei 1800 min. |

Nur originale Bauteile (MTU) und die entsprechenden Ersatzteile (MTU) verwenden, um die Lebensdauer des Motors zu verlängern.

Benutzerspezifische Hinweise zur Wartung des Motors.  
Nur von MTU genehmigte Ersatzteile verwenden.  
Nur von MTU genehmigte Ersatzteile verwenden.  
Nur von MTU genehmigte Ersatzteile verwenden.  
Nur von MTU genehmigte Ersatzteile verwenden.







**Hinweise für Betrieb und Wartung**

Motor typ  Ventilspiel bei  $^{\circ}\text{C}$

Motornummer  Einlaß (mm)  Auslaß (mm)

Startfüllung  Vorwärmtemperatur  $^{\circ}\text{C}$  min.

Einspritz-Förderbeginn  $^{\circ}\text{C}$   Öltemperatur  $^{\circ}\text{C}$  max.

Düsenöffnungsdruck (bar)  Kühlwassertemperatur  $^{\circ}\text{C}$  max.

Düsentyp  Einbauschräglage, längs  $^{\circ}$

Zündfolge

Alle weiteren technischen Daten sind der Betriebsanleitung zu entnehmen  
 MTU Motoren- und Turbinen-Union Friedrichshafen GmbH

**Betriebsstoffe nur nach MTU-Vorschrift Nr. 1061**

Only use Fluids and Lubricants as per MTU Specification No.1061

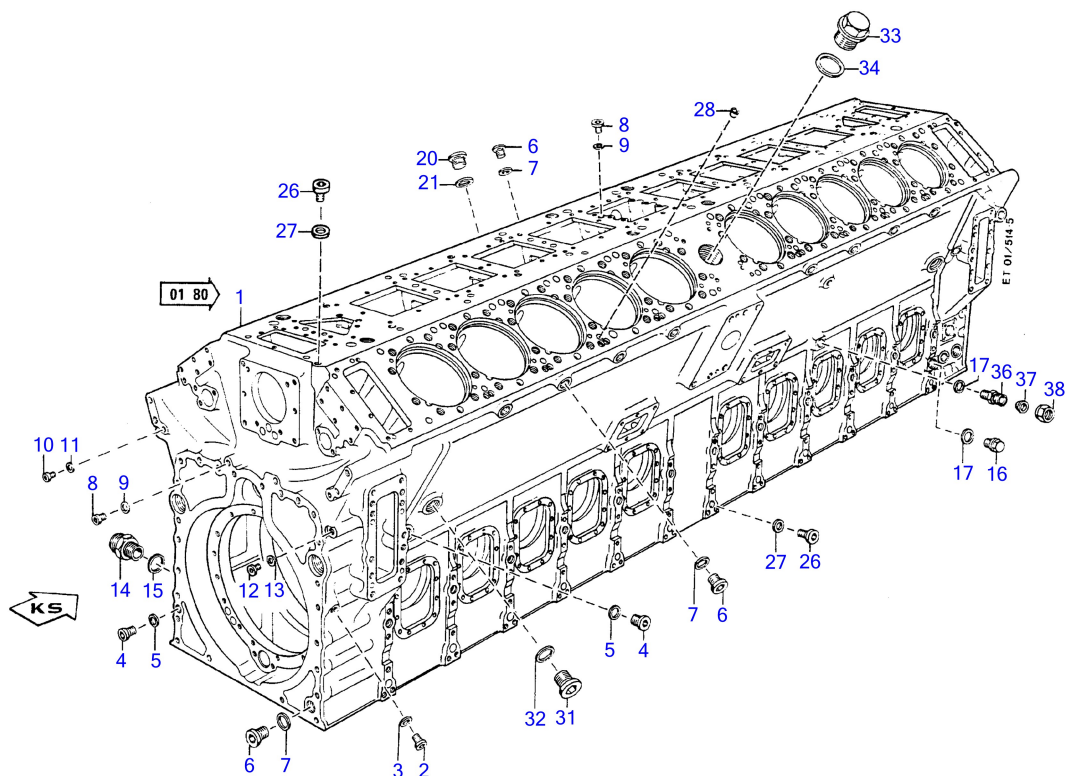
Utiliser les ingrédients prescrits par Publication MTU No. 1061

Utilizar solamente materias de servicio seg. Publ. MTU No.1061

Só utilizar agentes de funcionamiento seg. Publ. MTU No. 1061

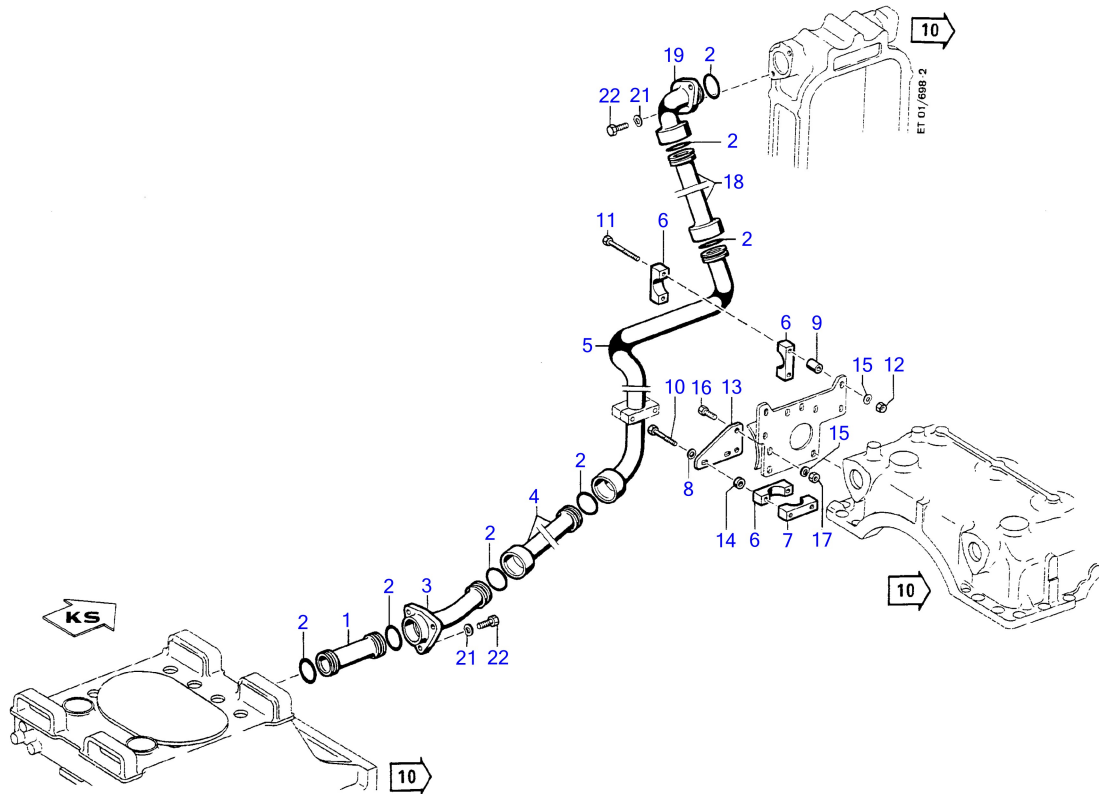
Impiegare solo materiali di consumo sec. Pubbl. MTU No. 1061

| POS. | PART NO.     | DESIGNATION       | QTY. | UNITS | FN | MEASUREMENT | STANDARD | DES. ADDITION |
|------|--------------|-------------------|------|-------|----|-------------|----------|---------------|
| 01   | 700425000050 | INSTRUCTION PLATE | 1    | EA    | NB | 40          |          |               |
| 02   | 700425000039 | INSTRUCTION PLATE | 1    | EA    |    | 36          |          |               |
| 03   | 700425000073 | INSTRUCTION PLATE | 1    | EA    | NB | 51          |          |               |

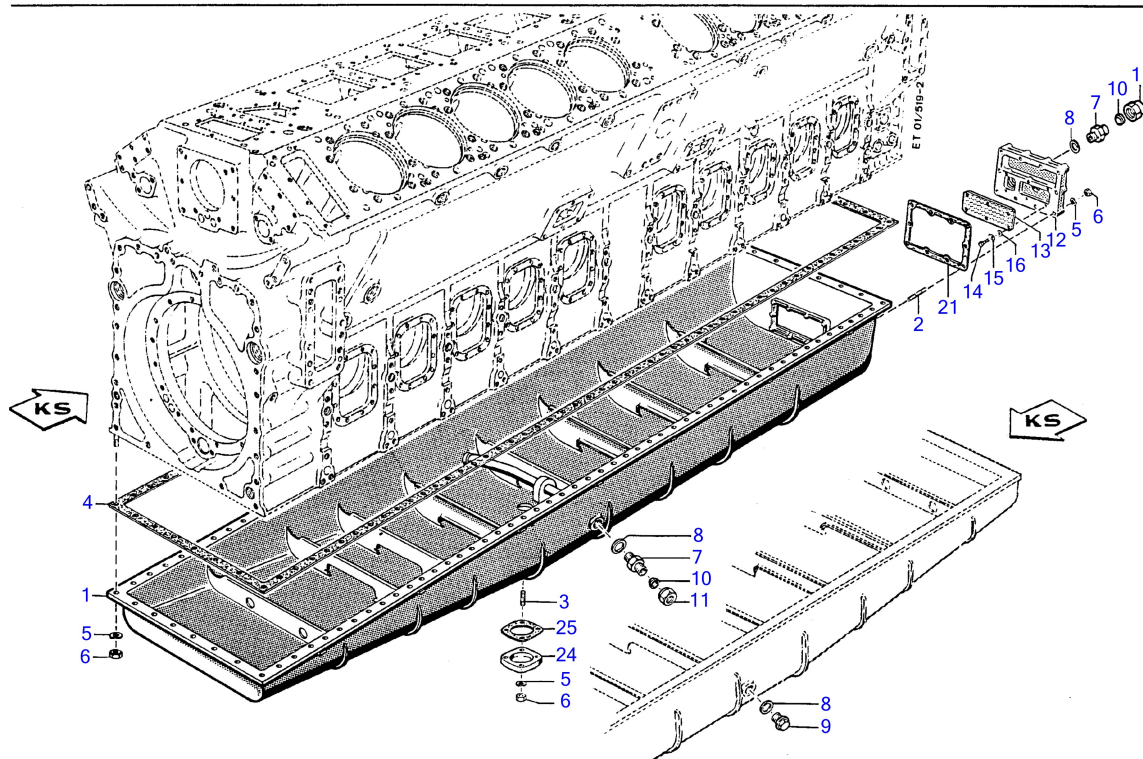


| POS. | PART NO.     | DESIGNATION       | QTY. | UNITS | FN | MEASUREMENT | STANDARD | DES. ADDITION |
|------|--------------|-------------------|------|-------|----|-------------|----------|---------------|
| 01   | 5620101105   | CRANKCASE         | 1    | EA    |    |             |          |               |
| 02   | 000908020000 | PLUG              | 4    | EA    |    | M 20 X 1,5  |          |               |
| 03   | 007603020101 | SEALING RING      | 4    | EA    |    | A 20 X 24   |          |               |
| 04   | 000908022001 | PLUG              | 6    | EA    |    | M 22 X 1,5  |          |               |
| 05   | 007603022102 | SEALING RING      | 6    | EA    |    | A 22 X 27   |          |               |
| 06   | 000908030002 | PLUG              | 21   | EA    |    | M 30 X 1,5  |          |               |
| 07   | 007603030101 | SEALING RING      | 21   | EA    |    | A 30 X 36   |          |               |
| 08   | 000908016001 | PLUG              | 5    | EA    |    | M 16 X 1,5  |          |               |
| 09   | 007603016105 | SEALING RING      | 5    | EA    |    | A 16 X 20   |          |               |
| 10   | 000908012003 | PLUG              | 2    | EA    |    | M 12 X 1,5  |          |               |
| 11   | 007603012111 | SEALING RING      | 2    | EA    |    | A 12 X 15,5 |          |               |
| 12   | 000908014001 | PLUG              | 2    | EA    |    | M 14 X 1,5  |          |               |
| 13   | 007603014102 | SEALING RING      | 2    | EA    |    | A 14 X 18   |          |               |
| 14   | 915013032000 | EXHAUST PIPE NECK | 1    | EA    |    | 32          |          |               |
| 15   | 007603038101 | SEALING RING      | 1    | EA    |    | A 38 X 44   |          |               |
| 16   | 000908026005 | PLUG              | 2    | EA    |    | M 26 X 1,5  |          |               |
| 17   | 007603026104 | SEALING RING      | 4    | EA    |    | A 26 X 31   |          |               |
| 20   | 000908036007 | PLUG              | 1    | EA    |    | M 36 X 1,5  |          |               |
| 21   | 007603036101 | SEALING RING      | 1    | EA    |    | A 36 X 42   |          |               |
| 26   | 000908018002 | PLUG              | 3    | EA    |    | M 18 X 1,5  |          |               |
| 27   | 007603018101 | SEALING RING      | 3    | EA    |    | A 18 X 22   |          |               |
| 28   | 700088020000 | PLUG              | 16   | EA    |    | 20          |          |               |
| 31   | 000908048004 | PLUG              | 4    | EA    |    | M 48 X 1,5  |          |               |

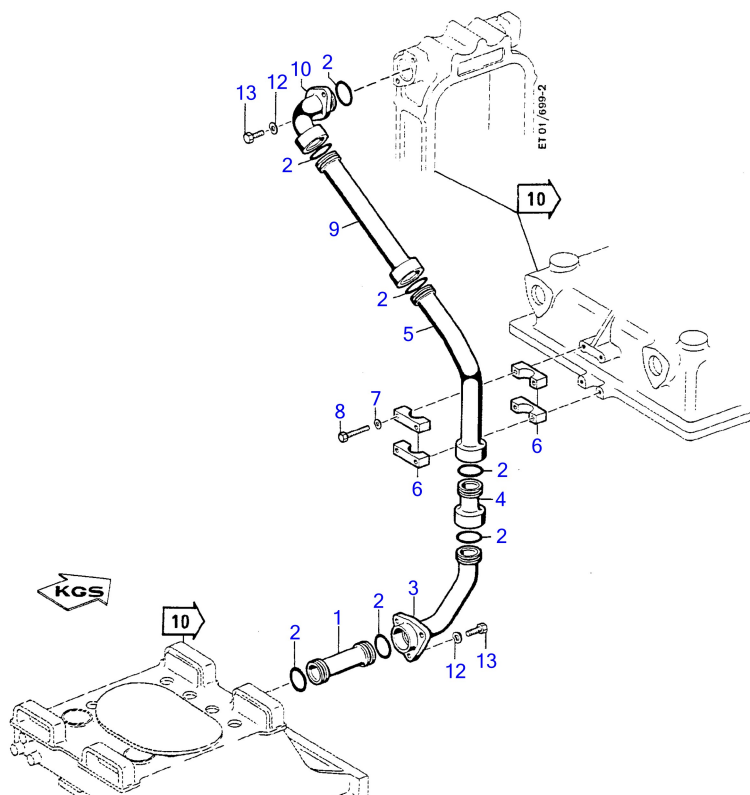
| POS. | PART NO.     | DESIGNATION     | QTY. | UNITS | FN | MEASUREMENT   | STANDARD | DES. ADDITION |
|------|--------------|-----------------|------|-------|----|---------------|----------|---------------|
| 32   | 007603048101 | SEALING RING    | 4    | EA    |    | A 48 X 55     |          |               |
| 33   | 000910064000 | PLUG            | 2    | EA    |    | M 64 X 2      |          |               |
| 34   | 007603064100 | SEALING RING    | 2    | EA    |    | A 64 X 72     |          |               |
| 36   | 915007025004 | REDUCTION UNION | 2    | EA    |    | 25 / 16       |          |               |
| 37   | 915005025002 | SEALING CONE    | 2    | EA    |    | 28            |          |               |
| 38   | 915001025004 | RIVET NUT       | 2    | EA    |    | 25 M 38 X 1,5 |          |               |



| POS. | PART NO.     | DESIGNATION     | QTY. | UNITS | FN | MEASUREMENT  | STANDARD | DES. ADDITION |
|------|--------------|-----------------|------|-------|----|--------------|----------|---------------|
| 01   | 5620100555   | FUEL PIPELINE   | 1    | EA    | S  |              |          |               |
| 02   | 700429045000 | O-RING          | 7    | EA    |    | 45 X 5       |          |               |
| 03   | 5620101755   | SUCTION PIPE    | 1    | EA    | S  |              |          |               |
| 04   | 5620101655   | FUEL PIPELINE   | 1    | EA    | S  |              |          |               |
| 05   | 5620101955   | FUEL PIPELINE   | 1    | EA    | S  |              |          |               |
| 06   | 700325042100 | PIPE HALF-CLAMP | 3    | EA    |    | 1/2 D 9 - 42 |          |               |
| 07   | 700325042200 | PIPE HALF-CLAMP | 1    | EA    |    | 1/2 G 8 - 42 |          |               |
| 08   | 000125008427 | COVER DISC      | 4    | EA    |    | A 8,4        |          |               |
| 09   | 5900942553   | SPACER SLEEVE   | 2    | EA    |    |              |          |               |
| 10   | 000931008181 | SCREW           | 2    | EA    |    | M 8 X 45     |          |               |
| 11   | 000931008207 | SCREW           | 2    | EA    |    | M 8 X 90     |          |               |
| 12   | 000934008008 | HEX NUT         | 2    | EA    |    | M 8          |          |               |
| 13   | 5620180140   | ALARM SWITCH    | 1    | EA    |    |              |          |               |
| 14   | 5509900440   | COVER DISC      | 2    | EA    |    |              |          |               |
| 15   | 000125010524 | COVER DISC      | 2    | EA    |    | A 10,5       |          |               |
| 16   | 000933010089 | SCREW           | 2    | EA    |    | M 10 X 20    |          |               |
| 17   | 000934010005 | HEX NUT         | 2    | EA    |    | M 10         |          |               |
| 18   | 5620102055   | FUEL PIPELINE   | 1    | EA    | S  |              |          |               |
| 19   | 5620101055   | SUCTION PIPE    | 1    | EA    | S  |              |          |               |
| 21   | 000137010202 | SPRING WASHER   | 5    | EA    |    | B 10         |          |               |
| 22   | 000933010035 | SCREW           | 5    | EA    |    | M 10 X 25    |          |               |



| POS. | PART NO.     | DESIGNATION      | QTY. | UNITS | FN | MEASUREMENT   | STANDARD | DES. ADDITION |
|------|--------------|------------------|------|-------|----|---------------|----------|---------------|
| 01   | 137E17741/3  | PLASTIC OIL PAN  | 1    | EA    | S  |               |          |               |
| 02   | 000939008079 | STUD             | 10   | EA    |    | M 8 X 50      |          |               |
| 03   | 000939008083 | STUD             | 8    | EA    |    | M 8 X 20      |          |               |
| 04   | 5620140022   | GASKET           | 1    | EA    |    |               |          |               |
| 05   | 000137008203 | SPRING WASHER    | 130  | EA    | B  | 8             |          |               |
| 06   | 000934008008 | HEX NUT          | 130  | EA    |    | M 8           |          |               |
| 07   | 915007032000 | REDUCTION UNION  | 2    | EA    |    | 32 / 20       |          |               |
| 08   | 007603030101 | SEALING RING     | 3    | EA    |    | A 30 X 36     |          |               |
| 09   | 000910030000 | PLUG             | 1    | EA    |    | M 30 X 1,5    |          |               |
| 10   | 915005032000 | SEALING CONE     | 2    | EA    |    | 32            |          |               |
| 11   | 915001032003 | RIVET NUT        | 2    | EA    |    | 32 M 45 X 1,5 |          |               |
| 12   | 5620100335   | END COVER        | 1    | EA    |    |               |          |               |
| 13   | 135E18023/1  | PERFORATED PLATE | 1    | EA    |    |               |          |               |
| 14   | 000933005004 | SCREW            | 6    | EA    |    | M 5 X 8       |          |               |
| 15   | 700291005300 | TAB WASHER       | 6    | EA    |    | 5,3           |          |               |
| 16   | 000125005322 | COVER DISC       | 6    | EA    |    | A 5,3         |          |               |
| 21   | 5600140080   | GASKET           | 1    | EA    |    |               |          |               |
| 24   | 135E18506/1  | END COVER        | 2    | EA    |    |               |          |               |
| 25   | 5600140180   | GASKET           | 2    | EA    |    |               |          |               |



| POS. | PART NO.     | DESIGNATION     | QTY. | UNITS | FN | MEASUREMENT  | STANDARD | DES. ADDITION |
|------|--------------|-----------------|------|-------|----|--------------|----------|---------------|
| 01   | 5620100555   | FUEL PIPELINE   | 1    | EA    | S  |              |          |               |
| 02   | 700429045000 | O-RING          | 7    | EA    |    | 45 X 5       |          |               |
| 03   | 5620101355   | SUCTION PIPE    | 1    | EA    | S  |              |          |               |
| 04   | 5620101455   | FUEL PIPELINE   | 1    | EA    | S  |              |          |               |
| 05   | 5620101555   | SUCTION PIPE    | 1    | EA    | S  |              |          |               |
| 06   | 700325042100 | PIPE HALF-CLAMP | 4    | EA    |    | 1/2 D 9 - 42 |          |               |
| 07   | 000137008203 | SPRING WASHER   | 4    | EA    |    | B 8          |          |               |
| 08   | 000933008242 | SCREW           | 4    | EA    |    | M 8 X 65     |          |               |
| 09   | 5620101855   | FUEL PIPELINE   | 1    | EA    | S  |              |          |               |
| 10   | 5620100755   | SUCTION PIPE    | 1    | EA    | S  |              |          |               |
| 12   | 000137010202 | SPRING WASHER   | 5    | EA    |    | B 10         |          |               |
| 13   | 000933010035 | SCREW           | 5    | EA    |    | M 10 X 25    |          |               |